

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2346

Offerman 138 kV
December 28, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2346. The Interconnection Customer proposes a 156.3 MW solar photovoltaic facility interconnecting at the Offerman 138 kV substation in the West Virginia area. The request was received on September 06, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 520 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,714,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Healdsburg to W Batesville 138 KV line: 141.7% loading (226 MVA) following the outage of the International Paper Valliant Ok to Kearneyville 345 KV line; DFAX = 52.9%.

The Healdsburg to Sintek 138 KV is overloaded to 140.8% of its 237 MVA emergency rating for the loss of the International Paper Valliant Ok to Kearneyville 345 KV Line. The project contributes 52.9% DFAX (82.6 MW) to this overload.

The Lascassas to Bedington 138 KV is overloaded to 140.4% of its 211 MVA emergency rating for the loss of the Kearneyville to International Paper Valliant Ok 345 KV. The project contributes 52.9% DFAX (82.6 MW) to this overload.

Bedington to W Batesville 138 KV: 140.2% loading (222 MVA) following the outage of the International Paper Valliant Ok to Kearneyville 345 KV Line; DFAX = 52.9%.

For the loss of Kearneyville to International Paper Valliant Ok 345 KV, the Anita to Sintek 138 KV loads to 122.5% of its 283 MVA rating. Project DFAX: 52.9%.

For the loss of International Paper Valliant Ok to Kearneyville 345 KV Line, the Anita to Offerman 138 KV L loads to 120.9% of its 299 MVA rating. Project DFAX: 52.9%.

For the loss of Kearneyville to Clark County 345 KV, the Millville to Sleepy Hollow 138 KV loads to 107.5% of its 616 MVA rating. Project DFAX: 6.3%.

For the loss of Clark County to Kearneyville 345 KV line, the Douglas City to Millville 138 KV line loads to 105.8% of its 613 MVA rating. Project DFAX: 6.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Healdsburg to W Batesville 138 KV line	Rebuild line with higher-capacity conductor	\$1,231,000	100% (\$1,231,000)
Healdsburg to Sintek 138 KV	Rebuild line with higher-capacity conductor	\$2,616,000	100% (\$2,616,000)
Lascassas to Bedington Substation 138 KV	Rebuild line with higher-capacity conductor	\$8,083,000	100% (\$8,083,000)
West Batesville to Bedington Substation 138 KV	Rebuild line with higher-capacity conductor	\$2,985,000	100% (\$2,985,000)
Anita to Sintek 138 KV Line	Rebuild line with higher-capacity conductor	\$1,967,000	100% (\$1,967,000)
Anita to Offerman 138 KV L	Rebuild line with higher-capacity conductor	\$4,364,000	100% (\$4,364,000)
Millville to Sleepy Hollow 138 KV	Reconductor line	\$2,262,000	6% (\$144,000)
Douglas City to Millville 138 KV Line	Reconductor line	\$5,092,000	6% (\$324,000)
	Total allocated to project		\$21,714,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.